In [1]:
```python
# importing pandas library
import pandas as pd
```

In [3]:
```python
# loading the dataset on 05-27-2020
covid_data = pd.read_csv(r"C:\Users\DELL\Documents\COVID-19-master\COVID-19-master\csse_covid_19_data\csse_covid_19_daily_reports\03-17-2020.csv")
```

In [5]:
```python
# Q1a: display first five rows
# covid_data.head() displays first five rows by default
covid_data.head()
```

Out[5]:

	Province/State	Country/Region	Last Update	Confirmed	Deaths	Recovered	Latitude	Longitude
0	Hubei	China	2020-03-17T11:53:10	67799	3111	56003	30.9756	112.2707
1	NaN	Italy	2020-03-17T18:33:02	31506	2503	2941	41.8719	12.5674
2	NaN	Iran	2020-03-17T15:13:09	16169	988	5389	32.4279	53.6880
3	NaN	Spain	2020-03-17T20:53:02	11748	533	1028	40.4637	-3.7492
4	NaN	Germany	2020-03-17T18:53:02	9257	24	67	51.1657	10.4515

In [7]:
```python
#Q1b: dataset information
covid_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 312 entries, 0 to 311
Data columns (total 8 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Province/State  154 non-null    object
 1   Country/Region  312 non-null    object
 2   Last Update     312 non-null    object
 3   Confirmed       312 non-null    int64
 4   Deaths          312 non-null    int64
 5   Recovered       312 non-null    int64
 6   Latitude        309 non-null    float64
 7   Longitude       309 non-null    float64
dtypes: float64(2), int64(3), object(3)
memory usage: 19.6+ KB
```

In [9]:
```python
#Q1c:check missing values
covid_data.isnull().sum()
```

Out[9]:
```
Province/State    158
Country/Region      0
Last Update         0
Confirmed           0
Deaths              0
Recovered           0
Latitude            3
Longitude           3
dtype: int64
```

In [13]:
```python
#Q2: latest number of confirmed, deaths, recovered and active cases of novel coronavirus (COVID-19) country wise

latestcovid_data = pd.read_csv(r"C:\Users\DELL\Documents\COVID-19-master\COVID-19-master\csse_covid_19_data\csse_covid_19_daily_reports\03-09-2023.csv")
latestcovid_data['Active'] = latestcovid_data['Confirmed'] - latestcovid_data['Deaths'] - latestcovid_data['Recovered']
latestcovid_data.groupby('Country_Region')[['Confirmed', 'Deaths', 'Recovered', 'Active']].sum()
```

Out[13]:

	Confirmed	Deaths	Recovered	Active
Country_Region				
Afghanistan	209451	7896	0.0	0.0
Albania	334457	3598	0.0	0.0
Algeria	271496	6881	0.0	0.0
Andorra	47890	165	0.0	0.0
Angola	105288	1933	0.0	0.0
...	...	...	...	...
West Bank and Gaza	703228	5708	0.0	0.0
Winter Olympics 2022	535	0	0.0	0.0
Yemen	11945	2159	0.0	0.0
Zambia	343135	4057	0.0	0.0
Zimbabwe	264276	5671	0.0	0.0

201 rows × 4 columns

In [15]:
```python
#Q3: latest number of confirmed, deaths and recovered people of novel coronavirus (COVID-19) country/region - province/state wise

latestcovid_data[['Country_Region', 'Province_State', 'Confirmed', 'Deaths', 'Recovered']]
```

Out[15]:

	Country_Region	Province_State	Confirmed	Deaths	Recovered
0	Afghanistan	NaN	209451	7896	NaN
1	Albania	NaN	334457	3598	NaN
2	Algeria	NaN	271496	6881	NaN
3	Andorra	NaN	47890	165	NaN
4	Angola	NaN	105288	1933	NaN
...	...	...	...	...	...
4011	West Bank and Gaza	NaN	703228	5708	NaN
4012	Winter Olympics 2022	NaN	535	0	NaN
4013	Yemen	NaN	11945	2159	NaN
4014	Zambia	NaN	343135	4057	NaN
4015	Zimbabwe	NaN	264276	5671	NaN

4016 rows × 5 columns

In [19]:

```
#Q4: Chinese province wise cases of confirmed, deaths and recovered cases of Novel Coronavirus(COVID-19)

china = covid_data[covid_data['Country/Region'] == 'China']
china.groupby('Province/State')[['Confirmed', 'Deaths', 'Recovered']].sum().reset_index()
china[['Province/State', 'Confirmed', 'Deaths', 'Recovered']]
```

Out[19]:

	Province/State	Confirmed	Deaths	Recovered
0	Hubei	67799	3111	56003
12	Guangdong	1364	8	1307
14	Henan	1273	22	1250
16	Zhejiang	1232	1	1216
19	Hunan	1018	4	1014
20	Anhui	990	6	984
22	Jiangxi	935	1	934
24	Shandong	761	7	746
27	Jiangsu	631	0	631
28	Chongqing	576	6	570
29	Sichuan	540	3	520
30	Heilongjiang	482	13	456
31	Beijing	456	8	369
36	Shanghai	358	3	325
40	Hebei	318	6	310
41	Fujian	296	1	295
45	Guangxi	253	2	248
46	Shaanxi	246	3	236
63	Yunnan	176	2	172
66	Hainan	168	6	161
67	Hong Kong	162	4	88
71	Guizhou	147	2	144
75	Tianjin	136	3	133
76	Gansu	133	2	91
77	Shanxi	133	0	133
79	Liaoning	125	1	120
89	Jilin	93	1	92
94	Xinjiang	76	3	73
95	Inner Mongolia	75	1	73
96	Ningxia	75	0	75
170	Qinghai	18	0	18
178	Macau	12	0	10
246	Tibet	1	0	1
275	Unknown	0	0	0

In [23]:

```
#Q5: latest country wise deaths cases of Novel Coronavirus(COVID-19)

latestcovid_data.groupby('Country_Region')['Deaths'].sum().reset_index()
```

Out[23]:

	Country_Region	Deaths
0	Afghanistan	7896
1	Albania	3598
2	Algeria	6881
3	Andorra	165
4	Angola	1933
...	...	...
196	West Bank and Gaza	5708
197	Winter Olympics 2022	0
198	Yemen	2159
199	Zambia	4057
200	Zimbabwe	5671

201 rows × 2 columns

In [29]:
```
# Q6: list of countries with no cases of Novel Coranavirus(COVID-19)

no_case_of_covid = covid_data.groupby('Country/Region')['Recovered'].sum().reset_index()
no_recovery = no_case_of_covid[no_case_of_covid['Recovered'] == 0]
print(no_recovery['Country/Region'].to_list())
```

```
['Albania', 'Antarctica', 'Antigua and Barbuda', 'Aruba', 'Barbados', 'Benin', 'Bhutan', 'Bolivia', 'Brunei', 'Bulgaria', 'Burkina Faso', 'Cameroon', 'Cen
tral African Republic', 'Chile', 'Congo (Brazzaville)', 'Congo (Kinshasa)', 'Costa Rica', 'Cuba', 'Cyprus', 'Dominican Republic', 'Ecuador', 'Equatorial G
uinea', 'Eswatini', 'Ethiopia', 'French Guiana', 'Gabon', 'Ghana', 'Greenland', 'Guadeloupe', 'Guam', 'Guatemala', 'Guernsey', 'Guinea', 'Guyana', 'Holy S
ee', 'Honduras', 'Iceland', 'Jersey', 'Kazakhstan', 'Kenya', 'Kiribati', 'Korea, North', 'Kosovo', 'Liberia', 'Liechtenstein', 'Luxembourg', 'Malaysia',
'Maldives', 'Martinique', 'Mauritania', 'Mayotte', 'Monaco', 'Mongolia', 'Montenegro', 'Namibia', 'Nauru', 'New Zealand', 'Nigeria', 'Palau', 'Panama', 'P
araguay', 'Puerto Rico', 'Republic of the Congo', 'Reunion', 'Rwanda', 'Saint Lucia', 'Saint Vincent and the Grenadines', 'Seychelles', 'Slovakia', 'Slove
nia', 'Somalia', 'South Africa', 'Sudan', 'Summer Olympics 2020', 'Suriname', 'Tanzania', 'The Bahamas', 'The Gambia', 'Togo', 'Tonga', 'Trinidad and Toba
go', 'Tunisia', 'Turkey', 'Tuvalu', 'Ukraine', 'Uruguay', 'Uzbekistan', 'Venezuela', 'Winter Olympics 2022', 'occupied Palestinian territory']
```

In [31]:
```
# Q7: list of countries with all cases of Novel Coranavirus(COVID-19) died

country_data = covid_data.groupby('Country/Region')[['Confirmed', 'Deaths', 'Recovered']].sum().reset_index()
all_died = country_data[(country_data['Confirmed'] == country_data['Deaths']) & (country_data['Recovered'] == 0)]
print(all_died['Country/Region'].to_list())
```

```
['Antarctica', 'Guernsey', 'Jersey', 'Kiribati', 'Korea, North', 'Malaysia', 'Nauru', 'Palau', 'Puerto Rico', 'Republic of the Congo', 'Sudan', 'Summer Ol
ympics 2020', 'Tonga', 'Tuvalu', 'Winter Olympics 2022', 'occupied Palestinian territory']
```

In [33]:
```
# Q8: list of countries with all cases of Novel Coranavirus(COVID-19) recovered

all_recovered = country_data[(country_data['Confirmed'] == country_data['Recovered']) & (country_data['Deaths'] == 0)]
print(all_recovered['Country/Region'].to_list())
```

```
['Antarctica', 'Guernsey', 'Jersey', 'Kiribati', 'Korea, North', 'Malaysia', 'Nauru', 'Nepal', 'Palau', 'Puerto Rico', 'Republic of the Congo', 'Summer Ol
ympics 2020', 'Tonga', 'Tuvalu', 'Winter Olympics 2022', 'occupied Palestinian territory']
```

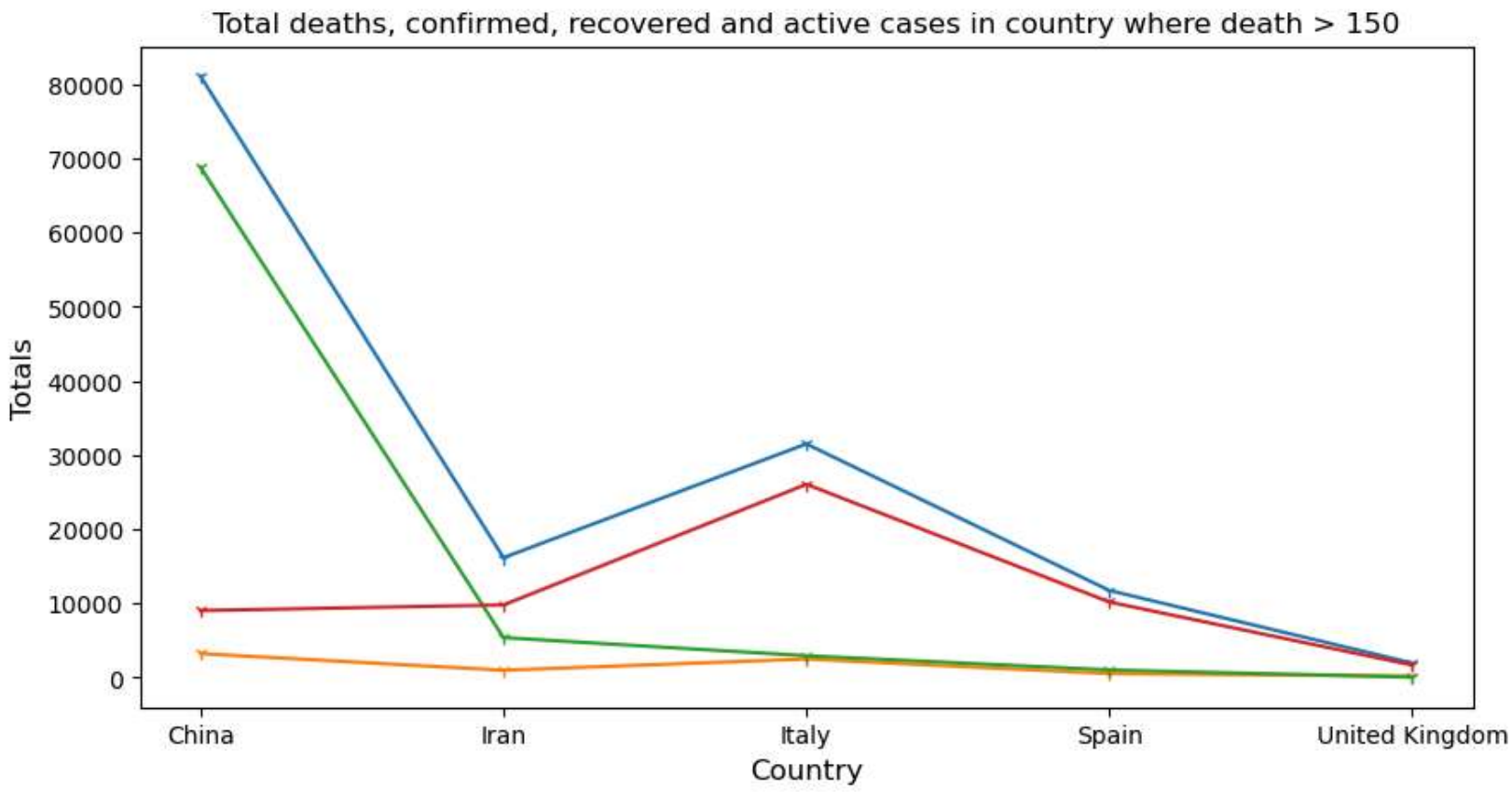
In [35]:
```
# Q9: top 10 countries data (last update, country/region, confirmed, deaths, recovered) of novel coronavirus(COVID-19)
top10country = covid_data.groupby('Country/Region')[['Last Update', 'Confirmed', 'Deaths', 'Recovered']].sum()
top10country.sort_values(by='Confirmed', ascending=False).head(10)
```

Out[35]:

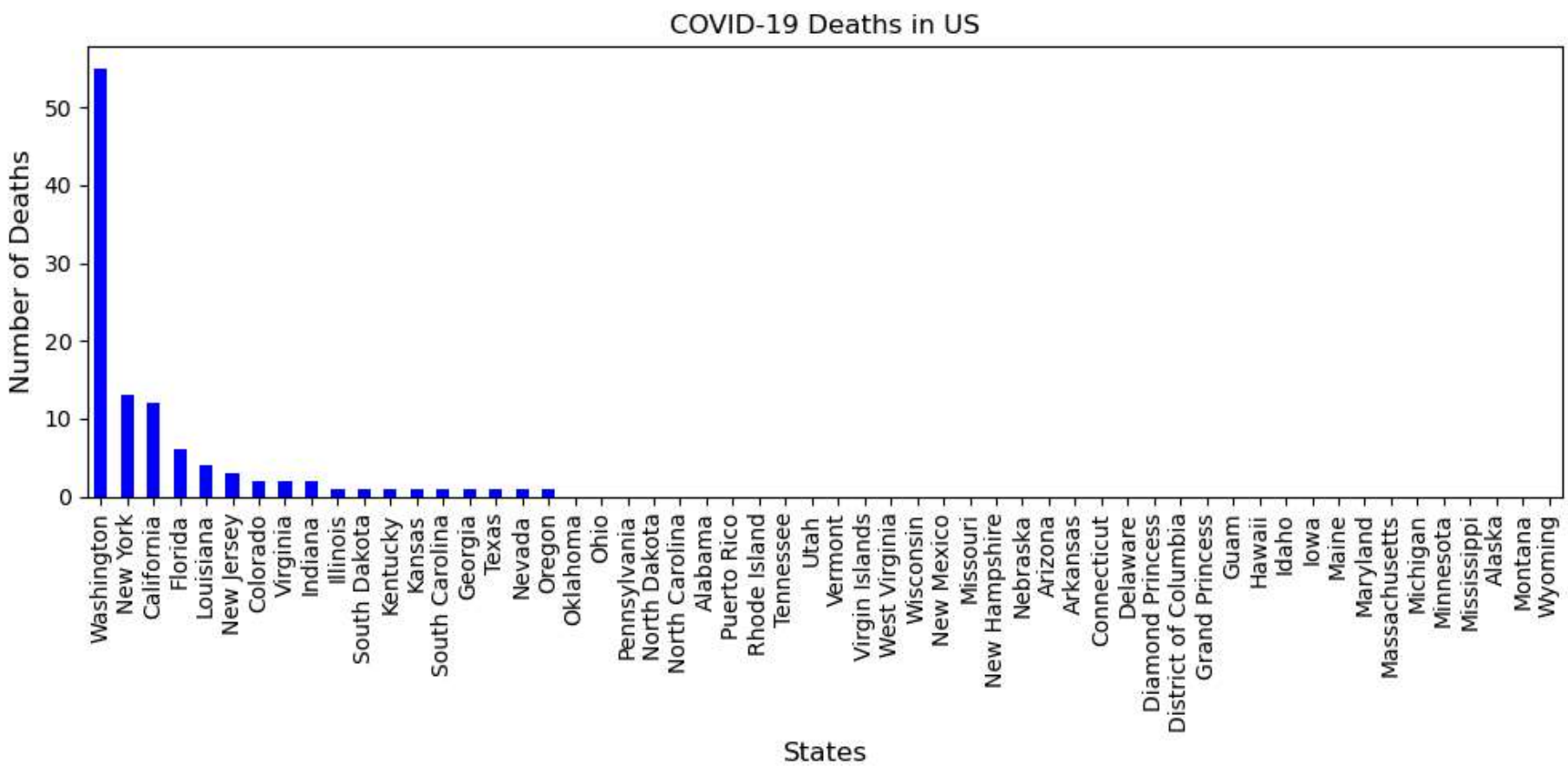
Country/Region	Last Update	Confirmed	Deaths	Recovered
China	2020-03-17T11:53:102020-03-17T01:53:032020-03-...	81058	3230	68798
Italy	2020-03-17T18:33:02	31506	2503	2941
Iran	2020-03-17T15:13:09	16169	988	5389
Spain	2020-03-17T20:53:02	11748	533	1028
Germany	2020-03-17T18:53:02	9257	24	67
Korea, South	2020-03-17T10:33:03	8320	81	1407
France	2020-03-17T19:13:082020-03-17T11:13:292020-03-...	7683	148	12
US	2020-03-17T22:53:032020-03-17T23:33:032020-03-...	6421	108	17
Switzerland	2020-03-17T16:33:04	2700	40	4
United Kingdom	2020-03-17T15:13:092020-03-16T14:38:452020-03-...	1954	217	53

In [37]:
```
# Q10: plot(lines) of total deaths, confirmed, recovered and active cases country wise where deaths is greater than 150

import matplotlib.pyplot as plt

countrywise = covid_data.groupby('Country/Region')[['Confirmed', 'Deaths', 'Recovered']].sum()
countrywise['Active'] = countrywise['Confirmed'] - countrywise['Deaths'] - countrywise['Recovered']
countrywise150 = countrywise[countrywise['Deaths'] > 150]
plt.figure(figsize=(10, 5))
plt.title('Total deaths, confirmed, recovered and active cases in country where death > 150', fontsize=12)
plt.xlabel('Country', fontsize=12)
plt.ylabel('Totals', fontsize=12)
plt.plot(countrywise150.index, countrywise150['Confirmed'], label='Confirmed', marker='1')
plt.plot(countrywise150.index, countrywise150['Deaths'], label='Deaths', marker='1')
plt.plot(countrywise150.index, countrywise150['Recovered'], label='Recovered', marker='1')
plt.plot(countrywise150.index, countrywise150['Active'], label='Active', marker='1')
```

Out[37]: [<matplotlib.lines.Line2D at 0x218dab44b60>]

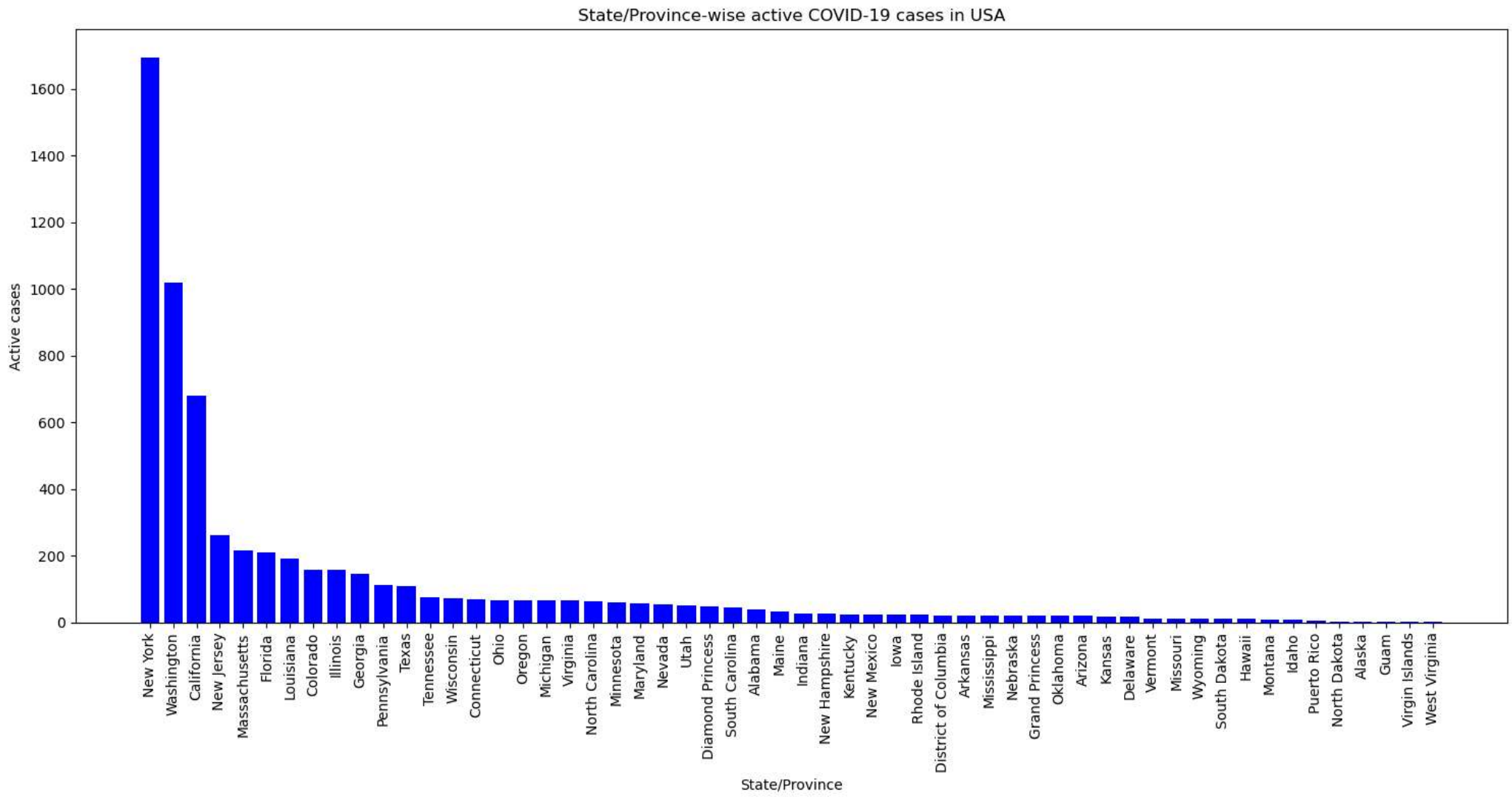


In [41]:
```
# Q11: visualize state/province wise death cases of novel coronavirus (COVID-19) in USA

usa = covid_data[covid_data['Country/Region'] == 'US']
statewise_death = usa.groupby('Province/State')['Deaths'].sum().sort_values(ascending=False)
plt.figure(figsize=(10, 5))
statewise_death.plot(kind='bar', color='blue')
plt.title('COVID-19 Deaths in US', fontsize=12)
plt.xlabel('States', fontsize=12)
plt.ylabel('Number of Deaths', fontsize=12)
plt.xticks(rotation=90)
plt.grid(axis='y', linestyle='', alpha=0.7)

plt.tight_layout()
plt.show()
```

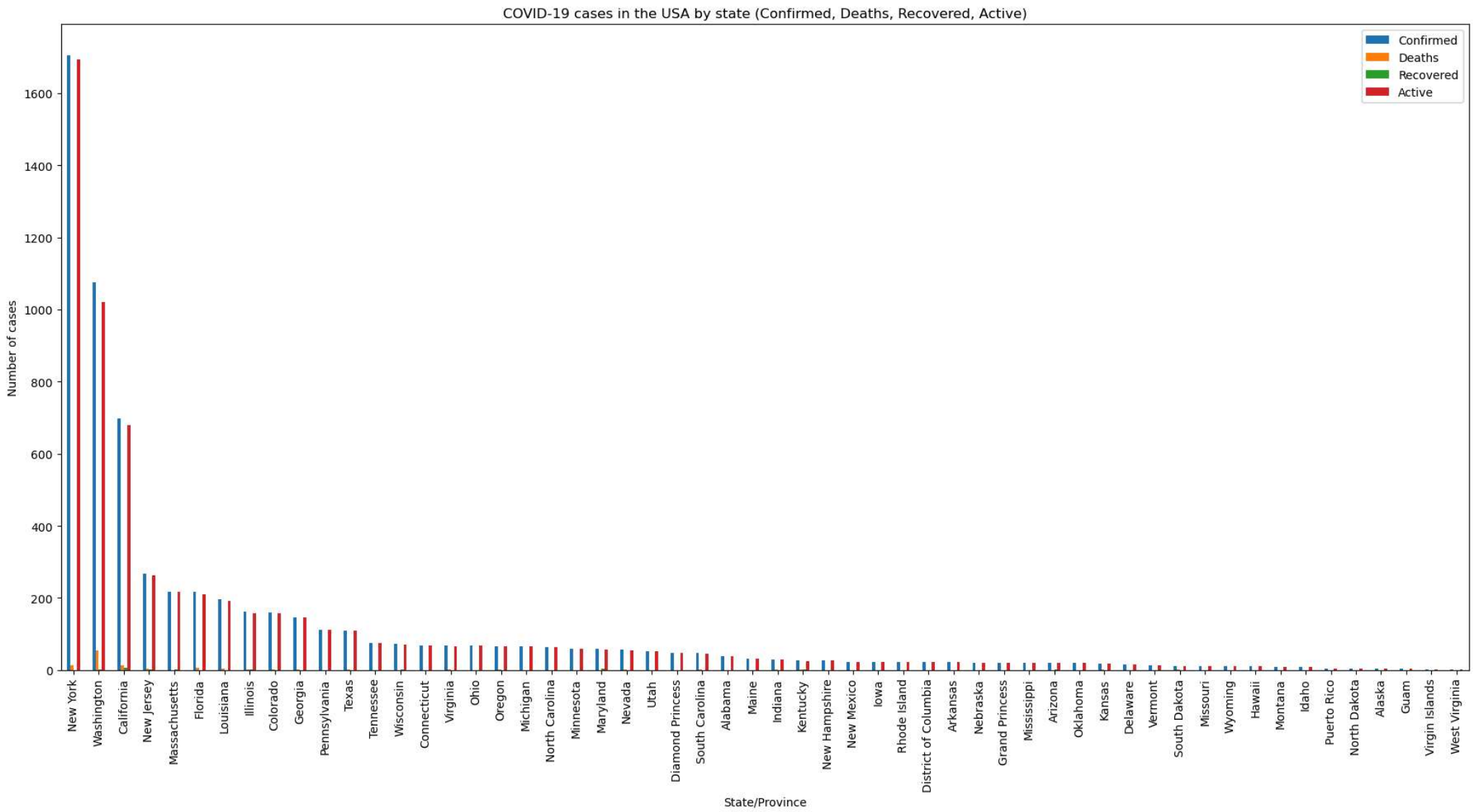


In [43]:
```
# Q12: visualize state/province wise active cases of novel coronavirus (COVID-19) in USA

#fill missing values to avoid errors in calculation
covid_data.fillna(0, inplace=True)

#filter for US
usa_df = covid_data[covid_data['Country/Region'] == 'US']

#group by state/pronvice and calculate totals
state_group = usa_df.groupby('Province/State')[['Confirmed', 'Deaths', 'Recovered']].sum()

#calculate active cases
state_group['Active'] = state_group['Confirmed'] - state_group['Deaths'] - state_group['Recovered']

#sort by active cases
state_group = state_group.sort_values(by='Active', ascending=False)

#plotting
plt.figure(figsize=(15, 8))
plt.bar(state_group.index, state_group['Active'], color='blue')
plt.title('State/Province-wise active COVID-19 cases in USA')
plt.xlabel('State/Province')
plt.ylabel('Active cases')
plt.xticks(rotation=90)
```

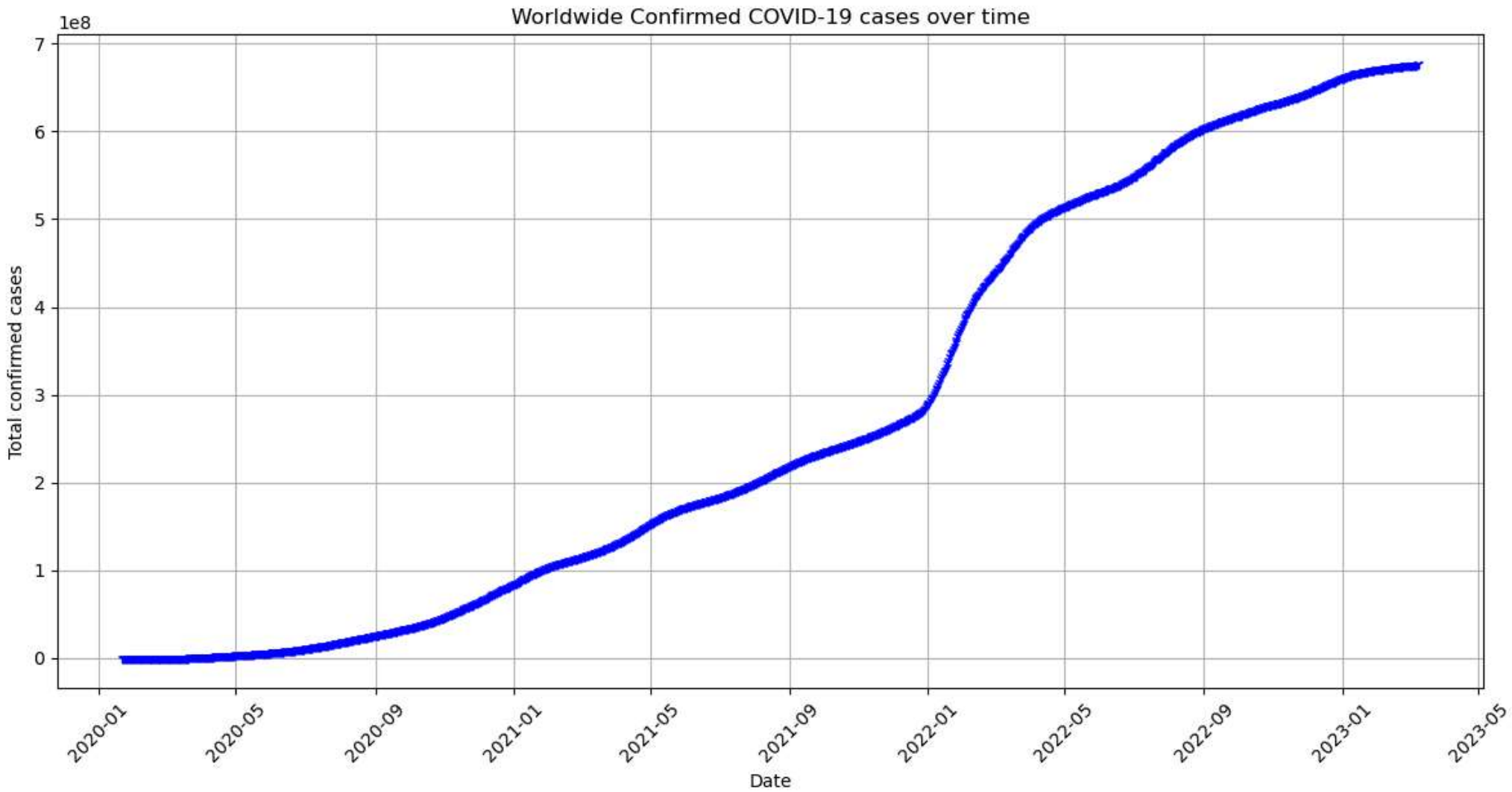
```
plt.tight_layout()
plt.show()
```



In [51]:
```
# Q13: visualize state/province wise combined number of comfirmed, death, recovered, active novel coronavirus (COVID-19) cases in USA

covid_data.fillna(0, inplace=True)

#group by state/province
grouped = usa.groupby('Province/State')[['Confirmed', 'Deaths', 'Recovered']].sum()

#calculate active cases
grouped['Active'] = grouped['Confirmed'] - grouped['Deaths'] - grouped['Recovered']

#sort by confirmed for better visuals
grouped = grouped.sort_values(by='Confirmed', ascending=False)

#plot
grouped[['Confirmed', 'Deaths', 'Recovered', 'Active']].plot(kind='bar', stacked=False, figsize=(18, 10))

plt.title('COVID-19 cases in the USA by state (Confirmed, Deaths, Recovered, Active)')
plt.xlabel('State/Province')
plt.ylabel('Number of cases')
plt.xticks(rotation=90)
plt.tight_layout()
plt.legend(loc='upper right')
plt.show()
```



In [45]:
```python
# Q14: visualize worldwide confirmed novel coronavirus (COVID-19) cases over time

import os

#set the folder path
folder_path = r"C:\Users\DELL\Documents\COVID-19-master\COVID-19-master\csse_covid_19_data\csse_covid_19_daily_reports"

#read and combine all csv
all_data = []
for file in os.listdir(folder_path):
    if file.endswith('.csv'):
        file_path = os.path.join(folder_path, file)
        df = pd.read_csv(file_path)
        df['Date'] = pd.to_datetime(file.replace('.csv', ''), format='%m-%d-%Y')
        all_data.append(df)

#concatenate into one dataframe
combined_df = pd.concat(all_data, ignore_index=True)

#group by date and sum confirmed cases
worldwide_confirmed = (combined_df.groupby('Date')['Confirmed'].sum().sort_index())

#plot
plt.figure(figsize=(12, 6))
plt.plot(worldwide_confirmed.index, worldwide_confirmed.values, marker='1', color='blue')
plt.title('Worldwide Confirmed COVID-19 cases over time')
plt.xlabel('Date')
plt.ylabel('Total confirmed cases')
plt.grid(True)
plt.tight_layout()
plt.xticks(rotation=45)
plt.show()
```



In [ ]: